

A Study of the Relationships between Service Failures and Customer Churn in a Telecommunications Environment

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Abstract--Telecommunications companies are facing increasing customer service pressure as they launch the triple-play bundle of voice, video, and internet access services. Firms recognize that providing quality customer service in the highly competitive telecommunications industry is critical to their success and delivering poor customer experiences puts customer relationships and revenues at risk.

One of the metrics used by telecommunications companies to determine their relationship with customers is "customer churn", the degree to which a telecommunication company's customers cancel or disconnect their service. An assumption in the industry is that customer churn is directly related to customer satisfaction. The purpose of this research is to identify customers with service failures and determine the propensity for a customer to disconnect based upon the frequency of a recent service failure reported and the success of repair. The research shows that customers who subscribe for the triple-play of voice, video and internet access are more likely to cancel all services from their provider after a service failure than are other customers. The results of the research suggest a strategy for identifying customers in advance who are most likely to disconnect and developing a service request response plan to encourage them to remain loyal customers.

I. INTRODUCTION

Telecommunication companies are facing more intensive customer service pressure than ever before as they launch the triple play bundle of services for Voice, Video, and Internet Access. Underlying the complexity of triple play delivery is the need to deliver a high-quality experience to customers, both when they are using the service and when they are receiving assistance from their service provider.

Cable industry veteran and Symmetricon board member Jim Chiddix describes why cable operators should measure customer experience: "Degraded video quality and other performance issues are real-world obstacles that can quickly increase cable operators' technical support costs, create customer churn, and dramatically impact revenue streams"[1].

Delivering a high-quality experience to customers is a primary goal for most service providers. The goal for telecommunications companies is to increase annual revenue per customer and customer loyalty, while reducing customer churn and leveraging overhead costs across their multi-service offering.

Providing quality customer service in the highly competitive telecommunications industry is critical to their success. "Research in the US has shown that 60 percent of DSL subscribers can expect to encounter problems during

IPTV installation. In a 2006 survey, also in the US, 34 percent of cable VoIP subscribers surveyed said that they had to have a technician visit their home to resolve a service issue within the first 90 days of installation"[2].

Delivering poor customer experiences puts customer relationships and revenues at risk. If a customer doesn't get their complaint resolved they "may not actually defect to a competitor for some time, but the likelihood of his defecting has suddenly increased, and the probability of his buying more things from you has just as suddenly declined. When this customer hangs up the phone... your company loses value at that moment"[5].

How the telecommunications provider reacts to a customer service issue is critical towards customer retention. "When a customer faces a service failure the service provider's reaction can potentially reinforce a strong customer bond and restore their satisfaction or it may exacerbate the situation and drive the customer to a competing firm"[6].

James Morehead the Vice President of product management and marketing at SupportSoft, Inc states that "In the triple play arena, customer churn is an even more painful scenario for service providers. If a triple play subscriber has a problem with just one service, that customer may cancel all services from that provider"[1].

In the most recent J.D. Power survey the top three factors driving overall customer satisfaction for telecommunication customers were Performance and Reliability (28%), Customer Service (22%), and Cost of Service (21%) [3].

Customer churn is a significant concern in the telecommunications industry. "One way to decrease churn in advance is to identify customers in advance who are at risk of churning and target an incentive to encourage them to stay"[4].

The purpose of this research is to identify customers with service failures and determine the propensity for a customer to disconnect based on the frequency of a recent service failure reported by the customer and the success of repair.

II. METHODOLOGY

The data used in this study is from monthly statistical reports of a national multi-system operator in the telecommunications industry over a 10 month period from January through October 2008.

Two approaches were used in this study. The first looked at the service experience of customers with a service failure, using a compilation of the telecommunications provider's

“phone survey statistics”⁸ of current customers with a service failure. This information identified the average number of visits to correct an issue, the success rate of the repair visit, and the percentage of customers who had multiple visits to correct the service failure. This information provided the ability to analyze the customer service experience of the average existing customer with a service failure.

The second approach looked at the frequency of customers who had disconnected their services following a service failure within a 30 day (monthly) reporting period, using empirical data from the telecommunications provider’s “billing system”[7].

The Venn diagram below (Fig. 1) shows data sets for total service calls requested, total service calls completed, and total customers that were disconnected within the same 30 day time period. A “service call requested” transaction occurs at the time a customer calls in to report a service related issue and a service call work order is entered into the billing system. A “service call completed” occurs when a service technician is dispatched to the residence or commercial property and corrects the service issue. The elements in light gray represent the intersection of the elements in the data sets for Total Service Calls Requested, Total Disconnected Customers, and Total Service Calls Completed.

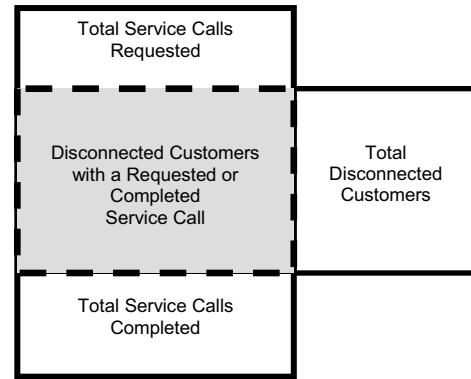


Fig. 1. Venn Diagram of Disconnected Customers, Total Services Calls

The disconnect data included information gathered from disconnecting customer exit interviews concerning the reason that they were requesting to disconnect their services. From this data set of disconnect reasons, those that were designated as non-preventable were excluded. These included those customers requesting to transfer their current services to a new address, those customers disconnected because of non-payment of bills, and deceased customers (Fig. 2).



Fig 2. Disconnected Customers Studied

Once the non-preventable work orders were removed from the data set of disconnected customers with a service call, the remaining data set was used to compare total requested and completed service calls.

The customer data also included the level of service that the customer subscribed to. This data allowed the opportunity to review the number of service failures requested and completed that resulted in a disconnection of service by type of customer based on level of service. Seven different customer groups were identified: those requesting video, data, or phone only; and those requesting multiple services (video/data, video/phone, data/phone, or video/data/phone).

III. RESULTS

A. Existing Customer Experience

The results from the first approach using a compilation of phone survey statistics of 525 current customers over the first

10 months of the year with a service failure allowed us to analyze and summarize the customer service experience of the average existing customer with a service failure. Based on the results of the surveys, it was found that 86% of the customers surveyed had their service failure resolved at the time of the phone survey. The results showed satisfaction levels as low as 80% and as high as 88% on a monthly basis over the first 10 months of 2008 (Table 1). The survey results also showed that, over the 10 month period of interest, 20% of the respondents had multiple visits regarding the service failure of one or more of their products or services. This percentage ranged as low as 10% and as high as 29% on a monthly basis during the period. The average home visits to resolve an issue was 1.4 and ranged between 1.02 and 1.6 over the period.

TABLE 1. SUMMARY OF HOME VISIT EXPERIENCES

Home Visit Experience	2008 YTD Jan – Oct Average
Base: Total Home Visit Respondents (n=)	105
Issue Resolved to Your Satisfaction (% Yes)	86%
Multiple Visits About Issue	20%
Average Visits Among Total Home Visits	1.4
Technician made sure services ran properly (% Yes)	88%

This information identified several reliability and customer service opportunities regarding customer satisfaction, customers requiring multiple visits to correct an issue, and resolving service failure issues on the first service repair visit. These opportunities will be discussed in a later section of this paper.

B. Disconnected Customer Experience

The second approach used data from the telecommunications firm's billing system. This data allowed us to analyze over 68,000 customer billing records to determine the frequency of customers who, over the 10 month period, had disconnected their services following a service failure reported within a 30 day (monthly) period prior to the disconnect. Table 2 summarizes the number of service call requests received and the number of disconnects monthly and over the 10 month period.

TABLE 2. SERVICE CALL AND DISCONNECT WORK ORDERS

	YTD Total	Monthly Average
Total service call requests	291,973	29,197
Total service calls completed	193,320	19,332
Total disconnects completed	107,182	10,718
Total savable disconnects (Excludes non-pays/transfers/deceased)	55,719	5,572

The total service calls requested during the period from January through October was 291,973, with an average of 29,197 service calls requested per month. The maximum number of service calls entered per month was 34,338 and the minimum was 23,925.

The total service calls completed during the period from January through October was 193,320 with an average of 19,332 service calls completed per month. The maximum number of service calls completed per month was 23,925 and the minimum was 17,119.

The total disconnects completed during the period from January through October was 107,182 with an average of 10,718 disconnects completed per month. The maximum number of disconnects per month was 11,866 and the minimum was 9,624.

The total potentially savable disconnects completed during the period from January through October was 55,719, with an average of 5,572 savable disconnects per month. The

maximum number of savable disconnects per month was 6,729 and the minimum was 4,616.

Service Call Requests on Disconnected Accounts - This section analyzed the number of service calls requested by those customers who ultimately discontinued the service agreement with the provider. The telecommunications provider was especially interested in this group of customers in order to understand the overall perception and experience that the customer has with their telecommunications provider. The customer who reports a service failure may experience a potentially negative perception of the provider even if the provider does not have to send a technician out to their home or place of business.

TABLE 3. SERVICE CALL REQUESTS AND TOTAL CUSTOMER DISCONNECTS

	YTD Total	Monthly Average
Total Service Call Requests on a Disconnected Account	3,166	317
Total Customers Disconnected based on Service Call Requests	2,577	258

Table 3 shows that, of the 2,577 customers requesting service who ultimately disconnected their service, 3,166 total service calls were received during the period from January through October, an average of 317 service calls requested per month. The maximum number of service calls requested on accounts that subsequently disconnected per month was 434 and the minimum was 242.

Table 4 shows the number of service calls requested per disconnected customer with a service call request. As can be seen by the data, 66.7% of the customer who ultimately discontinued service made only one service call request while over 96% of the customers who ultimately disconnected made only 3 or less service call requests. Obviously, the data suggests that the response to the first service call request is critical to "saving" the customer; the greatest opportunity to save a customer is within the first 3 service call requests.

The total number of customers who disconnected with a service call requested during the same month for the period from January through October was 2,577 with an average of 258 customers disconnecting per month. The maximum number of customers who disconnected with a service call requested per month was 332 and the minimum was 202.

The following exhibit (Table 5) shows the actual number of customers who disconnect sorted by the number of service requests per customer. As can be seen by the data below, 98% (104,605) of those accounts disconnected did so without requesting a service call, and therefore are not impacted by service request or response. Again, for those who do request service, a positive response to the first service request is seen to be essential.

TABLE 4. SERVICE CALLS REQUESTED PER DISCONNECTED CUSTOMER

Service calls requested per disconnected customer	Frequency of actual service call requests per disconnected customer frequency category	Relative frequency	Cumulative frequency of actual service call requests by disconnected customer frequency category	Cumulative Relative Frequency
0	0	0%	0	0%
1	2,113	66.74%	2,113	66.74%
2	754	23.82%	2,867	90.56%
3	186	5.87%	3,053	96.43%
4	64	2.02%	3,117	98.45%
5	35	1.11%	3,152	99.56%
6	6	0.19%	3,158	99.75%
7	0	0.00%	3,158	99.75%
8	8	0.25%	3,166	100.00%
Total	3,166			

TABLE 5. NUMBER OF DISCONNECTED CUSTOMERS BY NUMBER OF SERVICE CALL REQUESTS

Disconnected accounts by the # of service calls Requested	Frequency of disconnected customers by service call requests	Relative frequency	Cumulative frequency of disconnected customers by service requests	Cumulative Relative Frequency
0 service call requested	104,605	98%	104,605	97.596%
1 service call requested	2,113	1.97%	106,718	99.567%
2 service calls requested	377	0.35%	107,095	99.919%
3 service calls requested	62	0.06%	107,157	99.977%
4 service calls requested	16	0.01%	107,173	99.992%
5 service calls requested	7	0.01%	107,180	99.998%
6 service calls requested	1	0.00%	107,181	99.999%
7 service calls requested	0	0.00%	107,181	99.999%
8 service calls requested	1	0.00%	107,182	100.000%
Total customers who disconnected	107,182			
Total customer who disconnected with a service call request	2,577			

Service Calls Completed on Discontinued Accounts - This section shows the number of service calls completed on those customers who ultimately discontinued. The total service calls completed on accounts that subsequently disconnected during the period from January through October was 1,788 with an average of 179 service calls completed per month (Table 6). The maximum number of service calls completed on accounts that subsequently disconnected per month was 257 and the minimum was 141.

Table 7 shows the number of service calls completed per disconnected customer. As can be seen the greatest

opportunity to save customer is with the first service call (84.5% of the total number of completed calls completed).

TABLE 6. SERVICE CALLS COMPLETED

	YTD	Monthly
	Total	Average
Total service calls completed on disconnected accounts	1,788	179
Total customers disconnected based on service calls completed	1,645	165

TABLE 7. TOTAL SERVICE CALLS COMPLETED PER DISCONNECTED CUSTOMER

Total number of service calls completed per disconnected customer	Frequency of actual service calls completed per disconnected customer frequency category	Relative frequency	Cumulative frequency of actual service calls completed by disconnected customer frequency category	Cumulative Relative Frequency'
0	0	0%	0	0%
1	1,510	84.45%	1,510	84.45%
2	258	14.43%	1,768	98.88%
3	15	0.84%	1,783	99.72%
4	0	0.00%	1,783	99.72%
5	5	0.28%	1,788	100.00%
Total	1,788			

The total number of customers who disconnected with a service call completed during the same month for the period from January through October was 1,645, with an average of 165 customers disconnecting per month. The maximum number of customers who disconnected with a service call completed per month was 230 and the minimum was 130.

Table 8 shows the actual number of customers who disconnected sorted by the number of completed service calls per customer. As can be seen by the data the greatest opportunity to save customers is within the first 2 categories. This is where 99.994% or 1,639 of the customers who disconnected with a service call completed were located.

TABLE 8. DISCONNECTS BY TOTAL COMPLETED SERVICE CALLS

Number of Customers who disconnected sorted by the number of TC's completed in 30 days	Frequency of disconnected customers by service calls completed	Relative frequency	Cumulative frequency of disconnected customers by service calls completed	Cumulative Relative Frequency'
Accts w/ 0 TC completed	105,537	98%	105,537	98.465%
Accts w/ 1 TC completed	1,510	1.41%	107,047	99.874%
Accts w/ 2 TC completed	129	0.12%	107,176	99.994%
Accts w/ 3 TC completed	5	0.00%	107,181	99.999%
Accts w/ 4 TC completed	0	0.00%	107,181	99.999%
Accts w/ 5 TC completed	1	0.00%	107,182	100.000%
Total disconnected customers	107,182			
Total customer who disconnected with a service call completed	1,645			

Level of Service Provided - The last section is an analysis of disconnects and the effect of service calls on customers by the level of service that the customer subscribed to.

The following chart (Figure 9) is an analysis of the number of service calls requested per disconnected customer with a service call request sorted by customer level of service. As can be seen by the data, the greatest opportunities to save customers are those with Video only (20.2%), Video/Data (20.5%), or Video/Data/Phone (41%). These three service categories accounted for 81.6% of the service calls by customers who disconnected.

TABLE 9. NUMBER OF SERVICE CALL REQUESTS BY SERVICE CATEGORY

Service calls requested resulting in disconnect by customer type	3,166	% by level of service
Video only	638	20.2%
Data only	318	10.0%
Phone only	4	0.1%
Video/Data	648	20.5%
Video/Phone	76	2.4%
Data/Phone	184	5.8%
Video/Data/Phone	1,298	41.0%
	3,166	

Table 10 shows the number of service calls completed per disconnected customer with a service call completion sorted by level of service. Again, the greatest opportunity to save customers are those with Video only (18.3%), Video/Data (21.4%), and Video/Data/Phone (40.4%) where 80.1% of the service calls fell for customers who disconnected with a service call completion.

TABLE 10. NUMBER OF SERVICE CALLS COMPLETED BY SERVICE CATEGORY

Service calls completed resulting in disconnect by customer type	1,788	% by level of service
Video only	327	18.3%
Data only	185	10.3%
Phone only	4	0.2%
Video/Data	382	21.4%
Video/Phone	52	2.9%
Data/Phone	115	6.4%
Video/Data/Phone	723	40.4%
	1,788	

C. Projected Disconnected Customers

Based upon the data analyzed in the preceding sections, an attempt was undertaken to predict the number of customers requesting service who would ultimately disconnect.

Disconnects per 1,000 Service Requests - The number of service calls requested per disconnected customer compared to the total number of service calls requested over the 10 month reporting period was examined. This analysis excludes disconnected customers that did not request a service call.

The ending cumulative relative frequency for disconnected customers by service call requests can be applied to any number of requested service calls to predict the number of disconnects for a given number of service calls. Assuming d represents the projected number of disconnects, then

$$d_s = \text{number of disconnects projected} \\ = \text{total sum frequency of disconnected customers by service call requests}$$

If:

$$R_s = \text{number of service calls requested, and}$$

f = relative frequency of disconnected customers by service call requests

then:

$$d_s = \sum R_s * f_i \quad (1)$$

over the range of number of service call requests, i .

The total number of service calls requested (from Table 2) was 291,973. Table 11 shows the relative frequency of disconnected customers by number of service call requests (from Table 3).

From Table 3, we see that the maximum number of services calls requested was $i = 8$. Thus, from this information:

$$\begin{aligned} d_s &= R_s * f_1 + R_s * f_2 + R_s * f_3 + R_s * f_4 + R_s * f_5 + R_s * f_6 + \\ &\quad R_s * f_7 + R_s * f_8 \\ &= R_s * (.7237\% + .1291\% + .0212\% + .0055\% + .0024\% \\ &\quad + .0003\% + .0000\% + .0003) \\ &= 291,973 * (.883\%) \\ d_s &= 2,577 \end{aligned}$$

TABLE 11. RELATIVE FREQUENCY OF DISCONNECT OCCURRENCES WITH SERVICE CALL REQUESTS

Disconnected accounts by the # of service calls Requested	Frequency of disconnected customers by service call requests	Relative frequency (f)	Cumulative frequency of disconnected customers by service requests	Cumulative Relative Frequency'
1 service call requested	2,113	0.7237%	2,113	0.724%
2 service calls requested	377	0.1291%	2,490	0.853%
3 service calls requested	62	0.0212%	2,552	0.874%
4 service calls requested	16	0.0055%	2,568	0.880%
5 service calls requested	7	0.0024%	2,575	0.882%
6 service calls requested	1	0.0003%	2,576	0.882%
7 service calls requested	0	0.0000%	2,576	0.882%
8 service calls requested	1	0.0003%	2,577	0.883%
Total number of service calls requested	291,973			

From this information, we can determine the expected number of projected disconnects per 1000 service calls requested. (Table 12). Specifically, for $R_s = 1000$:

$$d_s = R_s * .883\% = 8.83$$

TABLE 12. EXPECTED DISCONNECTS PER EACH 1000 SERVICE CALL REQUESTS

Propensity of customers to disconnect per 1000 service calls requested	
1 service call requested	7.24
2 service calls requested	8.53
3 service calls requested	8.74
4 service calls requested	8.80
5 service calls requested	8.82
6 service calls requested	8.82
7 service calls requested	8.82
8 service calls requested	8.83

Disconnects per 1,000 Service Calls Completed - In a similar manner, the expected number of disconnects as a function of service calls completed, d_c , can be calculated.

Utilizing the same calculation method as described above:

$d_c = \sum R_c * g_i$ over the range of number of completed service calls ($i = 1$ through 5)

where:

d_c = number of disconnects projected
= total sum frequency of disconnected customers by completed calls

R_c = number of completed service calls, and

g = relative frequency of disconnected customers by completed service calls

Utilizing the data given in Figure 8, the relative frequency g of disconnected customers as a function of the number of completed service calls was calculated.

$$\begin{aligned} d_c &= R_c * g_1 + R_c * g_2 + R_c * g_3 + R_c * g_4 + R_c * g_5 \\ &= R_c * (.781\% + .067\% + .003\% + .000\% + .001\%) \\ d_c &= 193,320 * .852\% = 1,646 \end{aligned}$$

The following chart (Table 13) shows the frequency of disconnected customers by service calls completed compared to the total number of service calls completed over the 10 month reporting period.

TABLE 13. RELATIVE FREQUENCY OF DISCONNECTED CUSTOMERS

Number of Customers who disconnected sorted by the number of TC's completed in 30 days	Frequency of disconnected customers by service calls completed	Relative frequency (g)	Cumulative frequency of actual service calls completed by disconnected customer frequency category	Cumulative Relative Frequency
Accts w/ 1 TC completed	1,510	0.781%	1,510	0.781%
Accts w/ 2 TC completed	129	0.067%	1,639	0.848%
Accts w/ 3 TC completed	5	0.003%	1,644	0.850%
Accts w/ 4 TC completed	0	0.000%	1,644	0.850%
Accts w/ 5 TC completed	1	0.001%	1,645	0.851%
Total number of service calls completed	193,320			

Finally, the projected number of disconnects per 1000 completed service calls is shown in the Table 14.

TABLE 14. DISCONNECT PROPENSITIES BY SERVICE CALLS COMPLETED

Propensity of customers with a service issue to disconnect per 1000 service calls completed	
Accts w/ 1 TC completed	7.81
Accts w/ 2 TC completed	8.48
Accts w/ 3 TC completed	8.50
Accts w/ 4 TC completed	8.50
Accts w/ 5 TC completed	8.51

D. Projected Loss of Revenue

Based upon the data analyzed in the preceding sections, an attempt was undertaken to predict the projected loss in

annual revenues for customers who had service related issues and who would ultimately disconnect their services.

Figure 3 shows the greatest opportunity to save customers and reduce customer churn is within the first 3 service call requests where 99.03% of the customers who disconnected their services with the provider. 2,113 customers disconnected after their first service call request, 754 after their second request, and 186 after their third request. Customers that request service calls and don't get their service request resolved in a timely manner are most likely to disconnect their services after the first 3 service requests. The projected loss of revenue from the first 2,113 customers with a monthly bill of approximately \$100 per month that disconnects their service after the first service request is \$211,300 per month and \$2,535,600 per year.

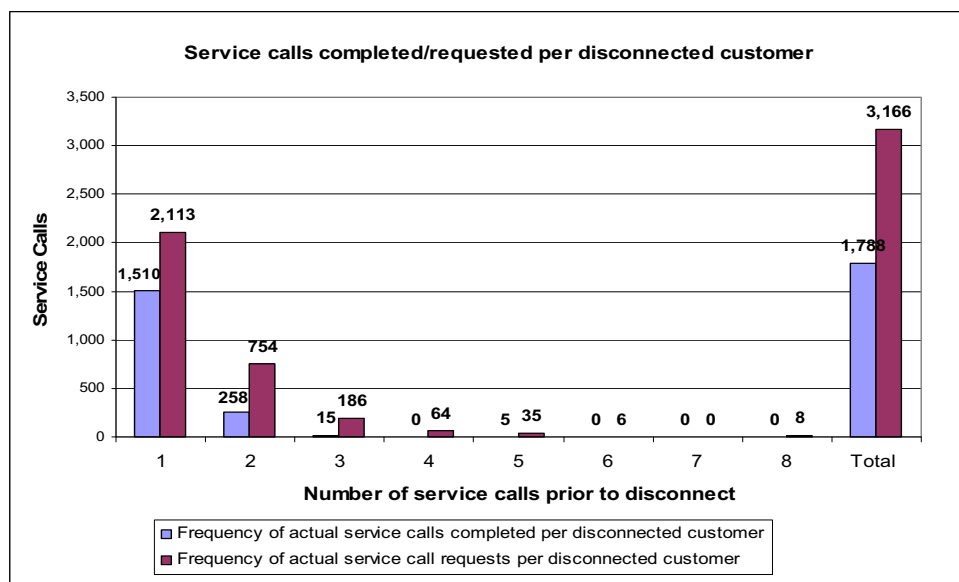


Figure 3. Service Calls Requested/Completed prior to Disconnect

Another opportunity to save customers is within the first 2 completed service calls. This is where 99.64% of the customers who had a service call completed and disconnected their services with their provider occurred. 1,510 customers disconnected after their first service call completion and 258 customers disconnected after their second completed service call. This clearly shows that in a competitive environment, customers expect to have reliable services. It also demonstrates that customers who disconnect with service issues expect to have a repair technician correct their service issues on the first service call. Most of the customers that

disconnected their services didn't give the service provider a second chance. The projected loss of revenue from the first 1,510 customers with a monthly bill of approximately \$100 per month that disconnect their service after the first service request is \$150,000 per month or \$1,812,000 per year.

An analysis of the customer billing system information identified the propensity for a customer to disconnect based on the frequency of a recent service failure being reported to be .88% and propensity for a customer to disconnect based on the frequency of a recent service failure with a completed service call work order is .85 %.

TABLE 15. DISCONNECTED CUSTOMERS WITH SERVICE ISSUES

	YTD	YTD Service Calls Resulting in a Disconnected Customer	Ratio of Disconnect Transactions to Service Transactions Completed	YTD Disconnected Customers With Service Issues	Ratio of Disconnected Customers to Service Transactions Completed	Disconnected Customers with Service Issues per 1000 Transactions
Service Call Request	291,973	3,166	1.08%	2,577	0.88%	8.83
Service Calls Completed	193,320	1,788	0.92%	1,645	0.85%	8.51
Variance	98,653	1,378	0.16%	932	0.03%	0.32

The variance in the number of customers who disconnected after a service call request was cancelled versus a service call completion was 932 customers (Table 15). The revenue per customer is around \$100 per month the revenue lost by these customers is approximately \$93,200 per month or \$1,118,400 per year.

The variance between a customer requesting a service call due to a service failure and the actual completion of a service calls was 98,653 service calls being cancelled.

The cost of rolling a truck is estimated at \$80 per customer visit for the telecommunications industry. The cost of rolling a truck on all 98,653 canceled service calls would be cost prohibitive in saving these 932 customers (\$7,892,240). Most of the canceled service calls were

associated with area service outages or area related service issues that could not be prevented if a service technician was rolled out to the specific residence.

There may be an opportunity in saving some of these customers by sending a service technician on specific types of customers. The division should prioritize dedicating their resources in a manner that would have the greatest propensity for saving the customers with the greatest revenue opportunity per month.

The analysis of customer's level of service demonstrates that the largest number of customers that are disconnecting with a service related issue is the "Triple Play" or Video/Data/Phone customer followed by Video/Data and Video only respectively.

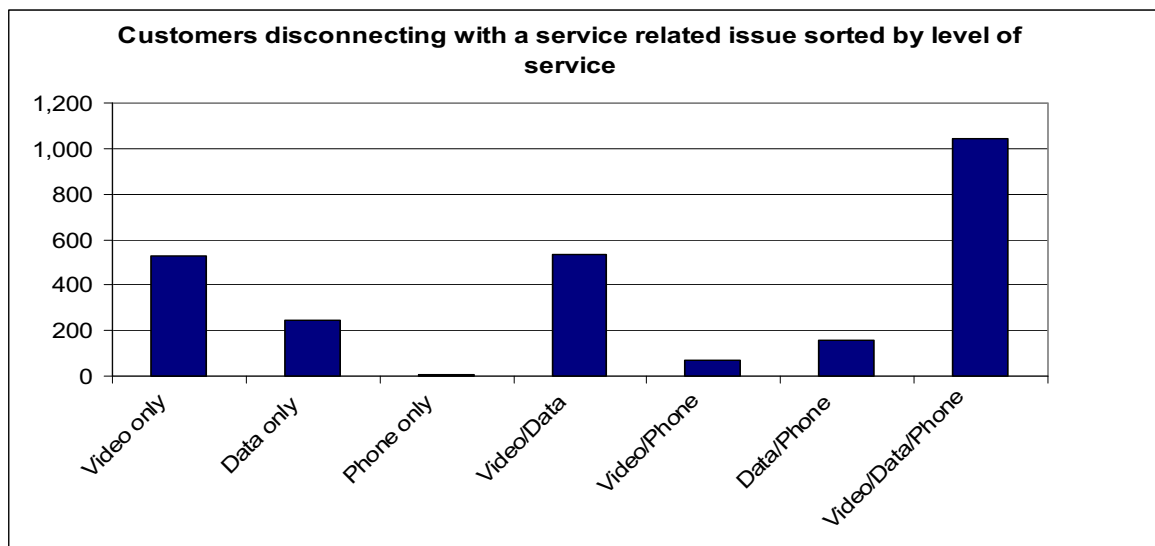


Figure 4. Disconnected Customers by Service Category

TABLE 16. DISCONNECTED VERSUS ACTIVE CUSTOMERS BY SERVICE CATEGORY

	Disconnect % of Customers with a Service Issue	Level of Service % for Active Customers	Variance (+%) variance is more tolerant versus (-%) is less tolerant)
Video only	20.6%	28.7%	8.2%
Data only	9.5%	8.0%	-1.5%
Phone only	0.2%	0.7%	0.5%
Video/Data	20.6%	20.7%	0.1%
Video/Phone	2.6%	4.8%	2.2%
Data/Phone	6.1%	1.6%	-4.6%
Video/Data/Phone	40.4%	35.4%	-5.0%

The exhibit above (Table 16) shows the distribution percentages of customers that disconnect with a service issue by service level compared to the distribution percentage of total active customers by service level. A positive variance demonstrates a greater tolerance to service related issues versus that of a negative variance which is less tolerant.

Based on the variances by category it appears that Video only customers are more tolerant to service issues since the distribution percentage for customers disconnecting for

service issues is less than the distribution percentage of the active Video only customers.

The "Triple Play", Data/Phone, and the Data only customers appear to be less tolerant to service issues since the distribution percentage for customers disconnecting for service issues is greater than the distribution percentage of the active customers in the same categories.

This observation reinforces the statement that a triple play subscriber who has a problem with just one service may

cancel all services from their provider. Based on these observations it is recommended to develop a retention strategy that focuses on trying to save your "Triple Play" customers since they appear to be the least tolerant category when it comes to disconnecting their services due to service failures and by the volume of services that they have they produce the greatest revenue per customer for the provider.

IV. CONCLUSION/RECOMMENDATIONS

The data and analysis in the report provides a method to identify customers in advance who are at risk of churning based on service call requests, service call, completion, and level of service when having a service failure.

In the highly competitive telecommunications market retaining every customer is essential to the overall profitability for the service provider. This analysis can be used as a road map that allows the telecommunications provider to target the customers that are in greatest jeopardy of disconnecting. This information also pinpoints the critical time in the customer's disconnect decision making process and allows the provider to develop an escalated service call response plan for the target customers. This also gives the provider the information to create an incentive plan to

encourage their customers to remain loyal to them prior to the customer making their decision to disconnect their services and going to a competitor.

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